RANIL K

Software Developer | Backend & Full-Stack Engineer

Madurai, India | ranilk691@gmail.com | +91-8825985792 | GitHub | LinkedIn | LeetCode

SUMMARY

Software Engineer with hands-on experience building backend systems, REST APIs, and full-stack web applications using Python, FastAPI, JavaScript/TypeScript, React, and Next.js. Strong foundation in data structures, algorithms, and object-oriented design, with additional experience integrating AI/LLM features (RAG pipelines, LLM APIs) into production-style applications. Collaborative and detail-oriented, focused on writing clean, scalable, well-tested code and shipping features end to end.

EDUCATION

B.E. Electronics and Communication Engineering
Velammal College of Engineering and Technology, MDU · 2021 – 2025 · CGPA: 8/10

Secondary and Senior Secondary Education
Dolphin Public School, MDU · 2019 – 2021 · 88% (10th), 90% (12th)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL
Backend & API Development: FastAPI, REST API Design, JWT Authentication, SQLAlchemy
Frontend Development: React, Next.js, Tailwind CSS, HTML5/CSS3, Streamlit
Databases: PostgreSQL, MongoDB
Core CS & Tools: Data Structures & Algorithms (100+ LeetCode), OOP, System Design, Git/GitHub,
AI & LLM Engineering: LangChain, LangGraph, Prompt Engineering, RAG, NLP, LLM

ATS Keywords: Python, JavaScript, TypeScript, React, Next.js, FastAPI, REST APIs, JWT Authentication, SQLAlchemy, PostgreSQL, MongoDB, System Design, Data Structures & Algorithms, OOP, Git, Full-Stack Development, Backend Development, Software Development, API Development, LangChain, RAG, LLM Integration

PROJECTS

FLASH – AI Interview Agent Platform · *Next.js · Streamlit · Python · FastAPI · Pinecone · RAG*

- Designed and built a full-stack interview automation platform, architecting a scalable FastAPI backend integrated with a Next.js and Streamlit frontend.
- Implemented custom RAG pipelines with Pinecone vector search to enable real-time, low-latency LLM-powered query resolution.

Intelligent Document Q&A API · *FastAPI · ChromaDB · PyMuPDF · Gemini API*

- Built a backend REST API that extracts and chunks PDF text, generates embeddings, and indexes them in ChromaDB for structured retrieval.
- Designed a grounded /ask endpoint integrating the Gemini API to return context-sourced answers with source metadata and graceful fallback handling.

Fleetfit · *Python · FastAPI · SQLAlchemy · PostgreSQL*

- Architected and deployed a modular, high-performance REST API backend for a gym management and fitness-tracking system.
- Modeled complex relational data using SQLAlchemy ORM across user profiles, workout tracking, and membership subscriptions; load-tested and optimized endpoints for strict data consistency.

INDEPENDENT PROJECTS & PRACTICE

Self-directed · Nov 2025 – OnGoing

- Built small front-end projects — TaskPulse (task manager), an e-commerce site with cart functionality, and an Expense Tracker — using HTML, CSS, and vanilla JavaScript, alongside core JS practice (calculator, tic-tac-toe) to reinforce logic building and DOM manipulation.
- Backend & AI exploration: practiced building backend services with FastAPI, PostgreSQL, and SQLAlchemy; built AI-focused experiments including an AI agent, a prompt playground, and a RAG app.

CERTIFICATIONS

Web Development Fundamentals · Python Programming · SQL · FastAPI — Udemy